CUSTOMER RETENTION & CHURN ANALYSIS

Future Interns – Data Science & Analytics Internship

Task 2

A data analytics project using Python and Power BI

Prepared for Internship Submission

1. Introduction

Customer churn is an important business problem because losing existing customers can reduce revenue and increase the cost of acquiring new customers. This project analyzes a telecommunications customer dataset to understand churn patterns and identify customer groups that may require retention efforts.

2. Project Objectives

The main objectives are to clean and analyze the customer dataset, calculate the overall churn rate, identify factors associated with churn, identify a practical high-risk customer segment, and present the findings through an interactive Power BI dashboard.

3. Dataset Description

The project uses the Telco Customer Churn dataset. It contains 7,043 customer records and fields covering customer information, tenure, services, contracts, payment methods, monthly charges, total charges and the Churn outcome. Churn = Yes means the customer has left the company, while Churn = No means the customer has stayed.

4. Tools and Technologies

Python, Pandas, NumPy and Matplotlib were used for data cleaning, analysis and visualization. Jupyter Notebook was used for the analytical workflow. Microsoft Power BI was used to create the interactive dashboard and present the business insights.

5. Data Cleaning

The TotalCharges column was converted from text to numeric format. Invalid or blank values were converted to missing values and rows with missing TotalCharges were removed. Missing-value checks and duplicate customer-ID checks were also performed before analysis.

6. Exploratory Data Analysis

The analysis examined overall churn, contract type, customer tenure, monthly charges, internet service, payment method and technical support. These comparisons were used to identify patterns that can support customer-retention decisions.

7. Key Findings

• The dataset contains 7,043 customers.
• 1,869 customers are churned.
• The overall churn rate is approximately 26.54%.
• Month-to-month customers show substantially higher churn than customers with one-year or two-year contracts.
• Customers with shorter tenure are an important retention group.
• Churn varies across internet-service types and payment methods.
• Customers without technical support show higher historical churn than customers with technical support.

8. High-Risk Customer Analysis

A high-risk customer should be an active customer who has not yet churned but has characteristics associated with higher historical churn. The selected practical risk segment is customers with a month-to-month contract and tenure of 12 months or less. For retention campaigns, only active customers (Churn = No) from this segment should be targeted.

9. Power BI Dashboard

The Power BI dashboard contains KPI cards for Total Customers, Churned Customers and Average Monthly Charges. It also contains visualizations for churn by contract type, churn by internet service, churned customers by payment method, churned customers by technical support, a churn gauge, and an interactive contract slicer.

10. Business Recommendations

1. Encourage suitable month-to-month customers to move to longer contracts using loyalty benefits or discounts.
2. Give additional attention to customers during their first 12 months.
3. Improve technical support and proactively contact customers who do not have support services.
4. Investigate service quality, pricing and customer experience for high-churn internet-service groups.
5. Review the reasons for higher churn among electronic-check users and promote convenient payment options.
6. Use customer-level risk indicators to prioritize retention campaigns.

11. Conclusion

The project demonstrates how data analytics can be used to understand customer churn and support retention decisions. Python provided the data-cleaning and exploratory-analysis workflow, while Power BI converted the results into an interactive business dashboard. Contract type, tenure and service-related factors provide useful signals for identifying customers who may need proactive retention attention.

12. Submission Checklist

■ Jupyter Notebook completed
■ Power BI dashboard completed
■ Dashboard screenshot added
■ Task 2 report PDF prepared
■ Project files/README prepared
■ Final submission uploaded to the internship portal

Key Project Metrics

Metric	Value
Total Customers	7,043
Churned Customers	1,869
Overall Churn Rate	26.54%

Average Monthly Charges	Approx. $64.76

Note: Add the final Power BI dashboard screenshot to the report before submitting if your submission requirements ask for visual evidence.